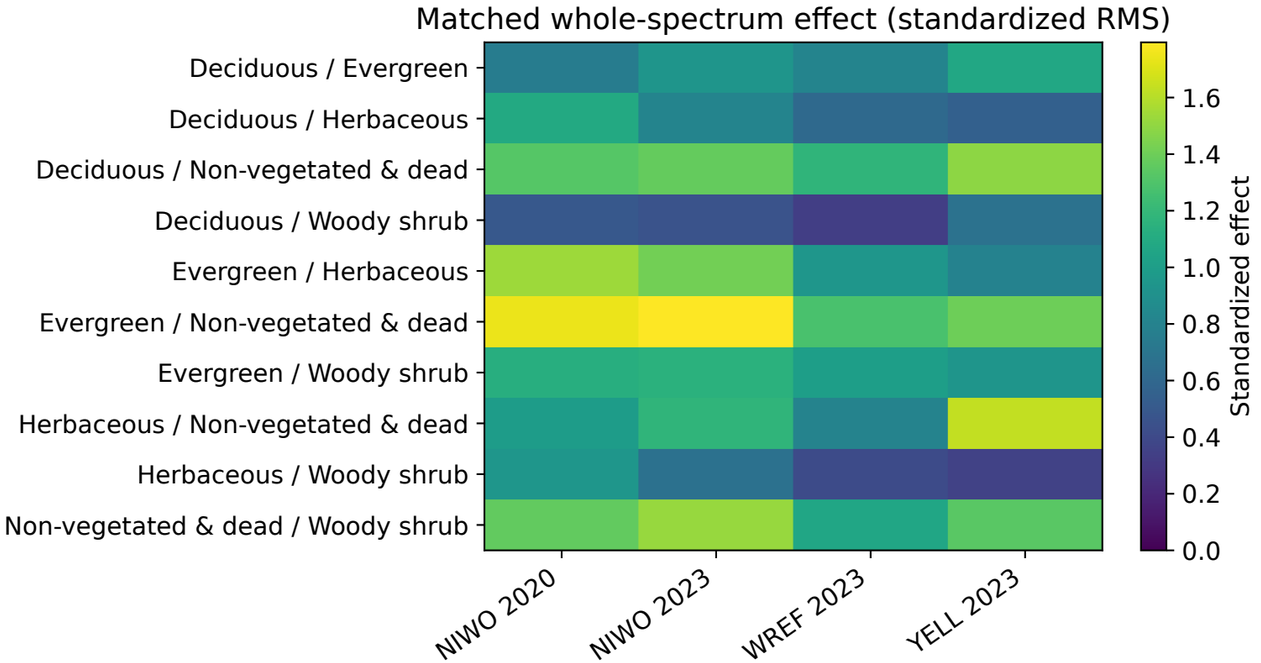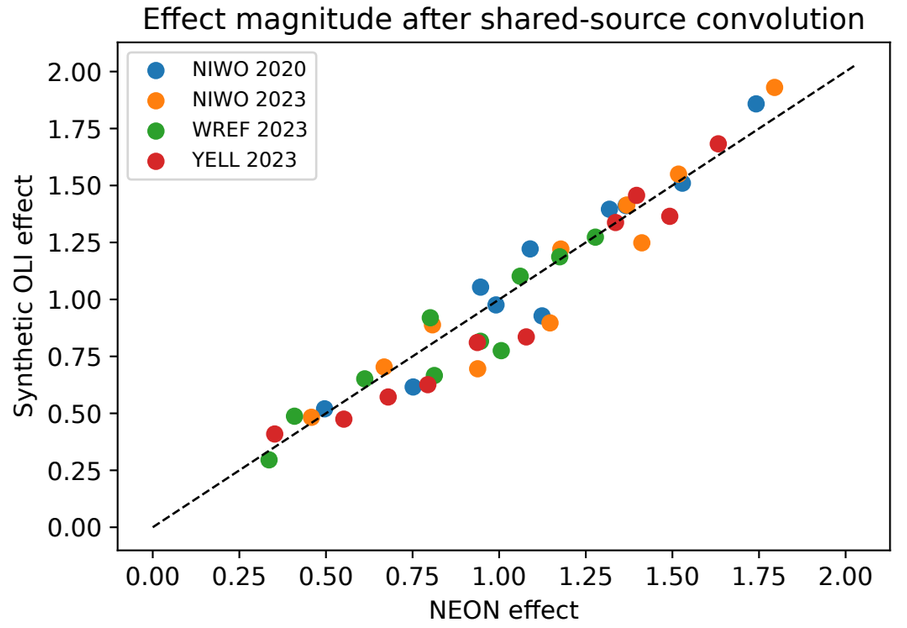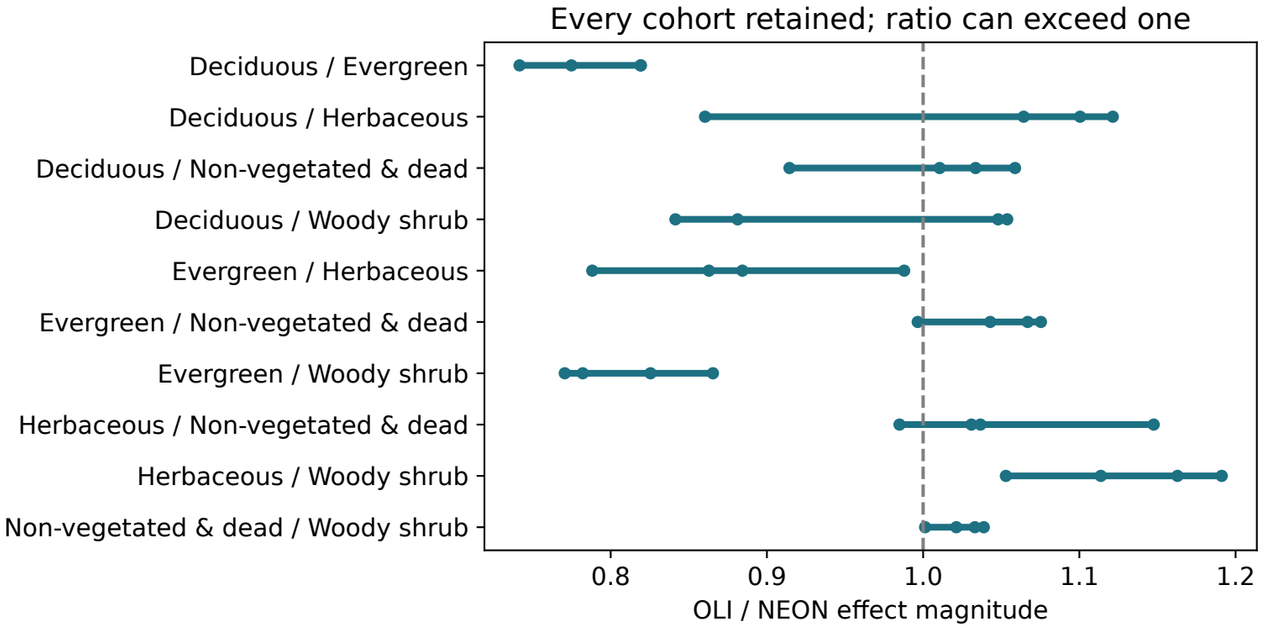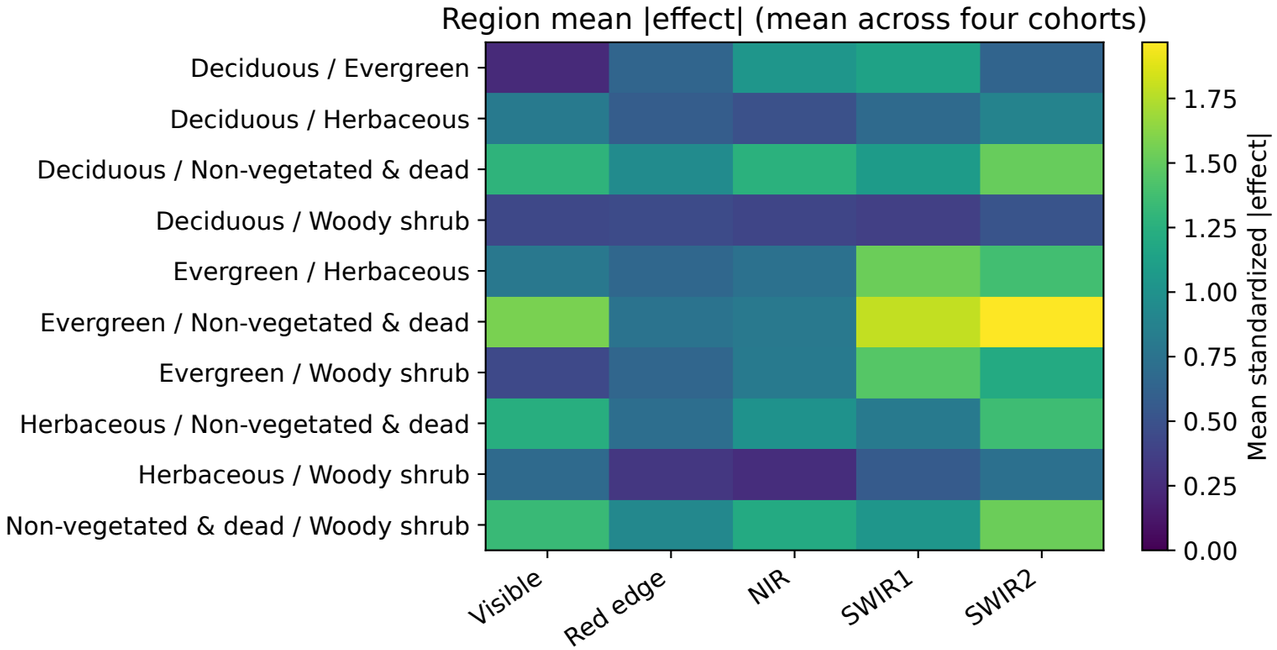Different spectra: effect size, statistical evidence and sensor retention